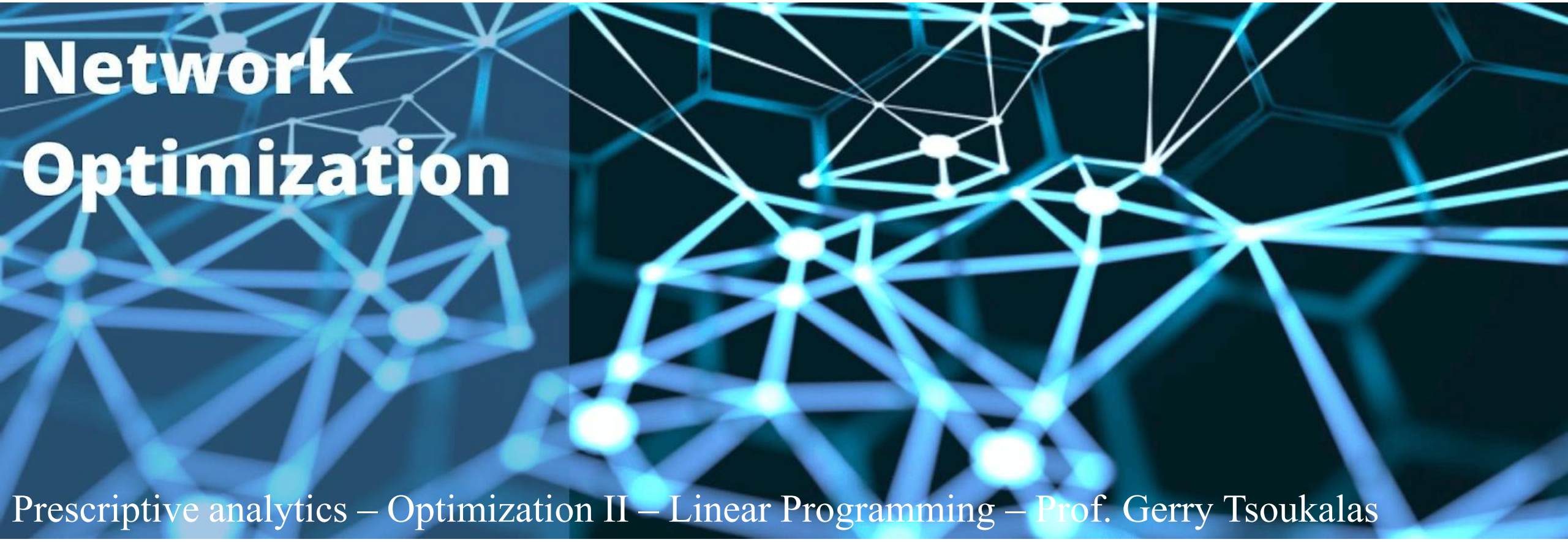

Network
Optimization
Prescriptive analytics – Optimization II – Linear Programming – Prof. Gerry Tsoukalas

Before we dive into networks…

- How do LPs find optimal solutions?

Recall: Domaine Paul Autard

- Has 12 hectares of vineyards in Chateauneuf du Pape
  - Each hectare produces about 25 hectoliters of wine
  - That's about 3,400 cases of wine in total
- The vineyards are planted with 3 types of grapes
  - 1,700 cases of Grenache
  - 1,200 cases of Syrah
  - 500 cases of Counoise
- The domaine produces two varieties of red wine
  - La Côte Ronde
    - » needs 0.5 cases of Grenache, 0.5 cases of Syrah, per case of wine
    - » will wholesale for 400€ per case
  - Domaine Paul Autard
    - » 0.6 cases Grenache, 0.2 cases Syrah, 0.2 cases Counoise, per case of wine
    - » will wholesale for 600€ per case
- How much of each wine should Paul Autard make to maximize sales?



Graphical Analysis of the LP

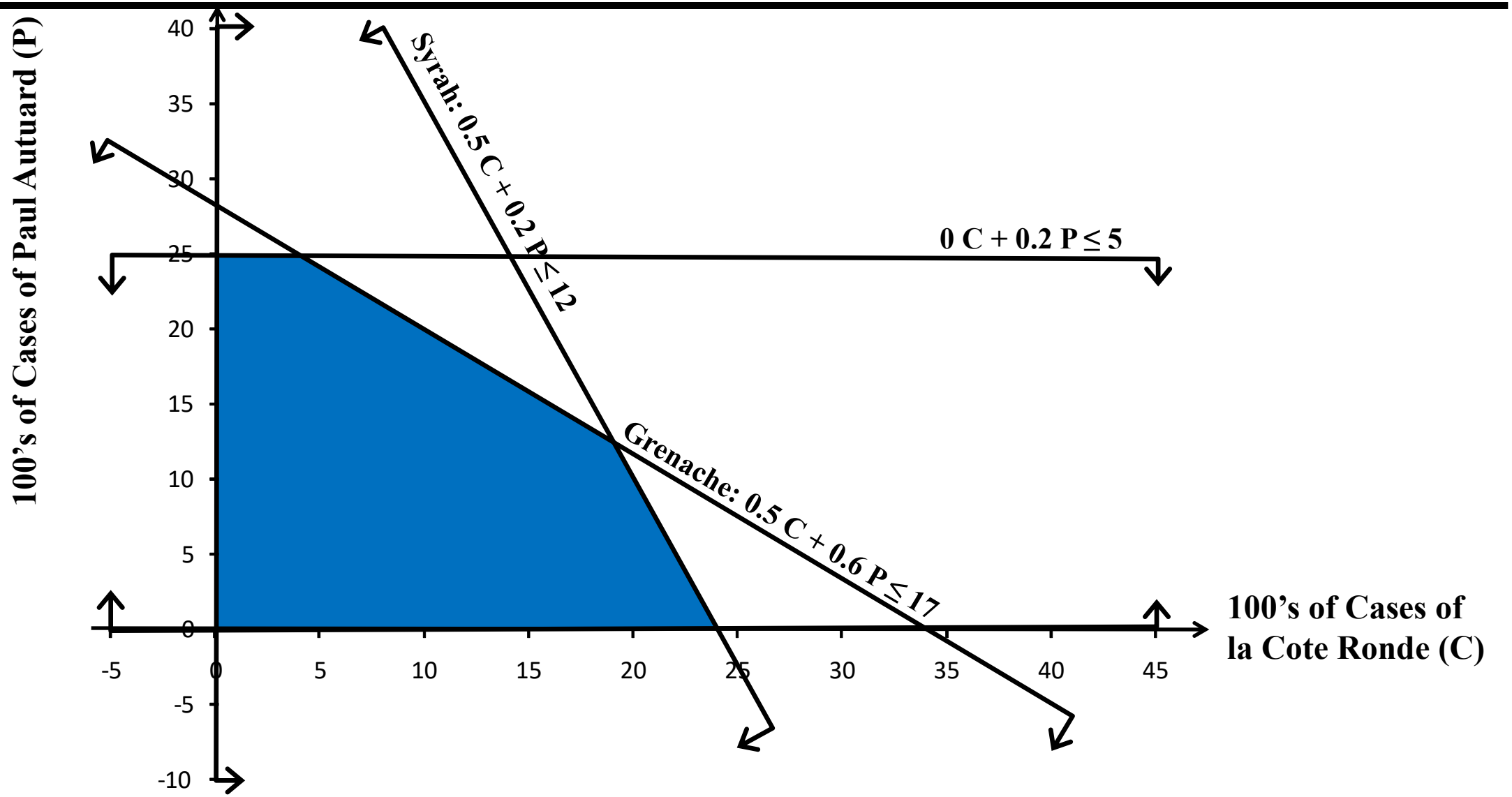
- Elements:
  - two decision variables
  - five inequality constraints
  - objective function

$$
\begin{aligned}
\max \quad & 40\,C + 60\,P \\
\text{subject to:} \quad & \\
0.5\,C + 0.6\,P &\le 17 \quad \text{(Grenache)} \\
0.5\,C + 0.2\,P &\le 12 \quad \text{(Syrah)} \\
0.2\,P &\le 5 \quad \text{(Counoise)} \\
1\,C &\ge 0 \quad \text{(non-negativity)} \\
1\,P &\ge 0 \quad \text{(non-negativity)}
\end{aligned}
$$

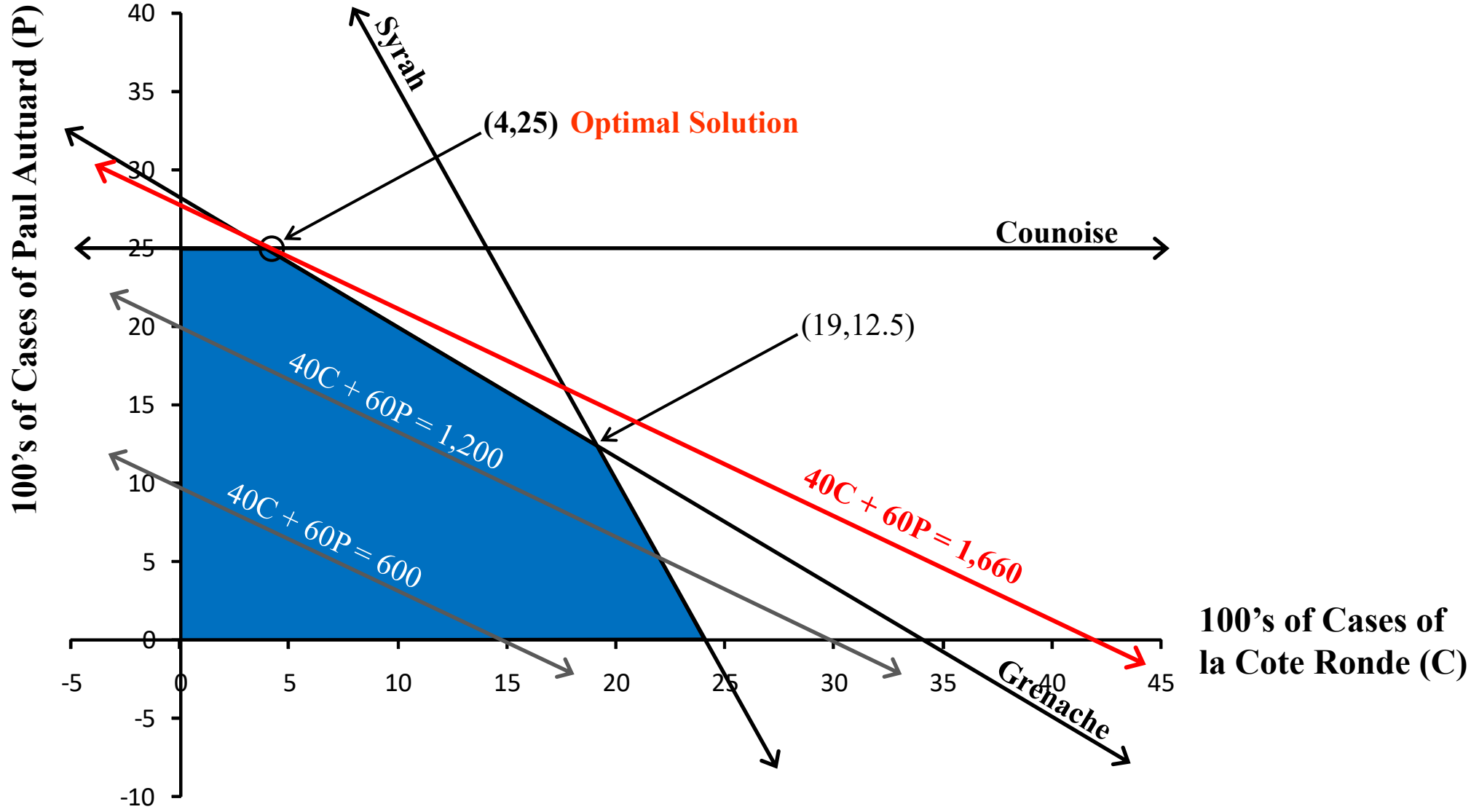
Graphical Analysis of the LP

Playground

The Feasible Region: Intersection of All the Constraints



An Optimal Solution Can Always be Found at a Vertex



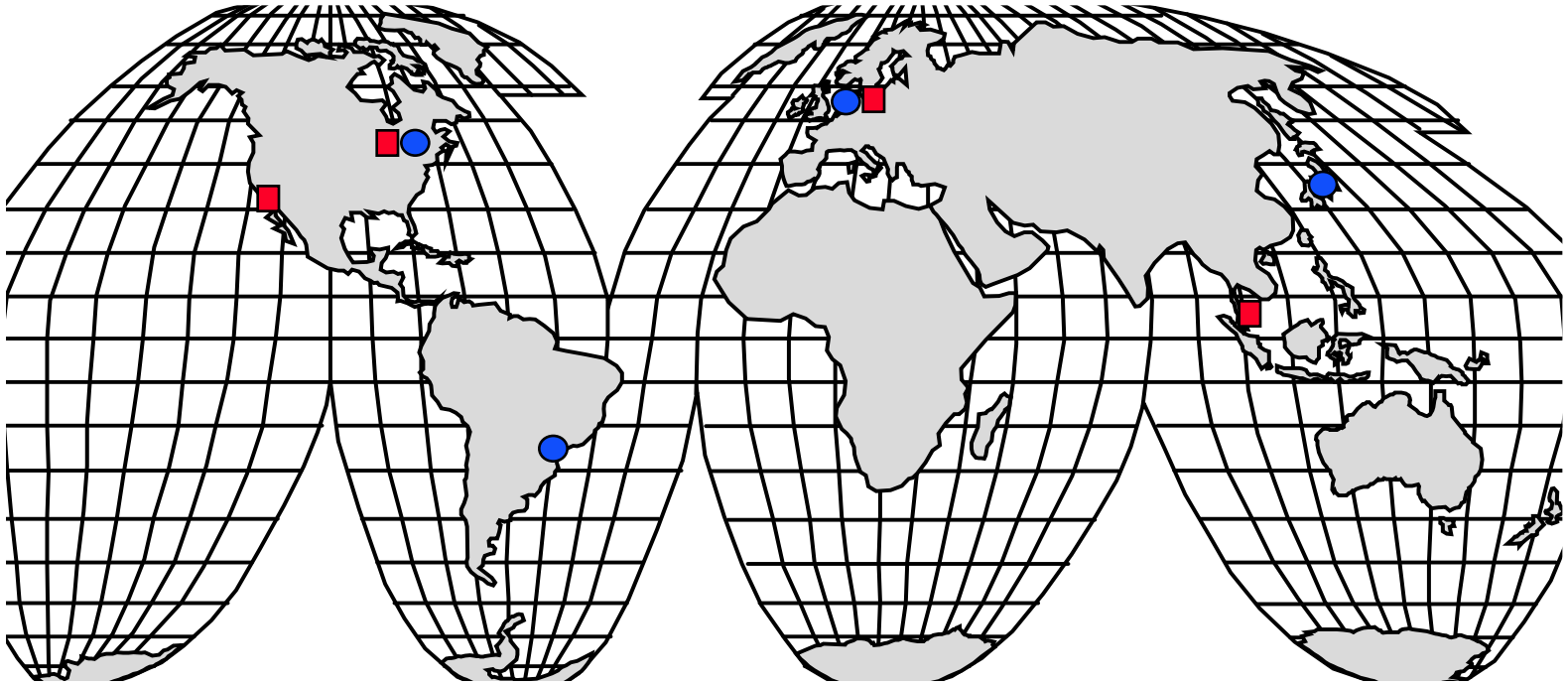
Summary

- Each decision variable defines one dimension of the problem space
  - two-variable problems can be plotted on a 2-D graph
  - pick an axis for each variable
- The constraints define the set of feasible solutions
  - each inequality defines a feasible *half-plane*
  - the problem's *feasible region* is the intersection of the half-planes
  - to draw a constraint: calculate the two intercepts; draw a line between them
- An optimal solution can always be found* at a *vertex* (corner point)
  - pick an arbitrary OFV and draw "test" line using intercepts
  - find the improving direction for the OFV; locate candidate optimal vertices
  - calculate the solution corresponding to each candidate vertex by solving the simultaneous equations of the vertex's *binding constraints*
  - compare OFVs, plugging solution coordinates into the objective function

*There may be no solution, or the problem may be unbounded.

Pfizer Vaccine Production/Transportation Problem

- A large multinational produces a popular vaccine in four dedicated plants, located in: **N**ewark; **L**os Angeles; **R**otterdam; and **K**uala Lumpur
- The vaccine is marketed worldwide, and the company has just instituted a regional sales organization with sales offices / distribution centers based in: **N**ewark; **R**otterdam; **S**ão Paulo; and **T**okyo



Prices and Costs Vary by Region and Plant

- Each sales / distribution center is responsible for a distinct region; the maximum ***demand*** and the average ***selling price*** for a region are given below.

Office	Newark	São Paulo	Rotterdam	Tokyo
Region	N. America	S. America	Eu / Af / ME	Asia
Market Price ($/ton)	55,500	61,100	57,800	62,650
Demands (tons/yr)	150	75	200	100

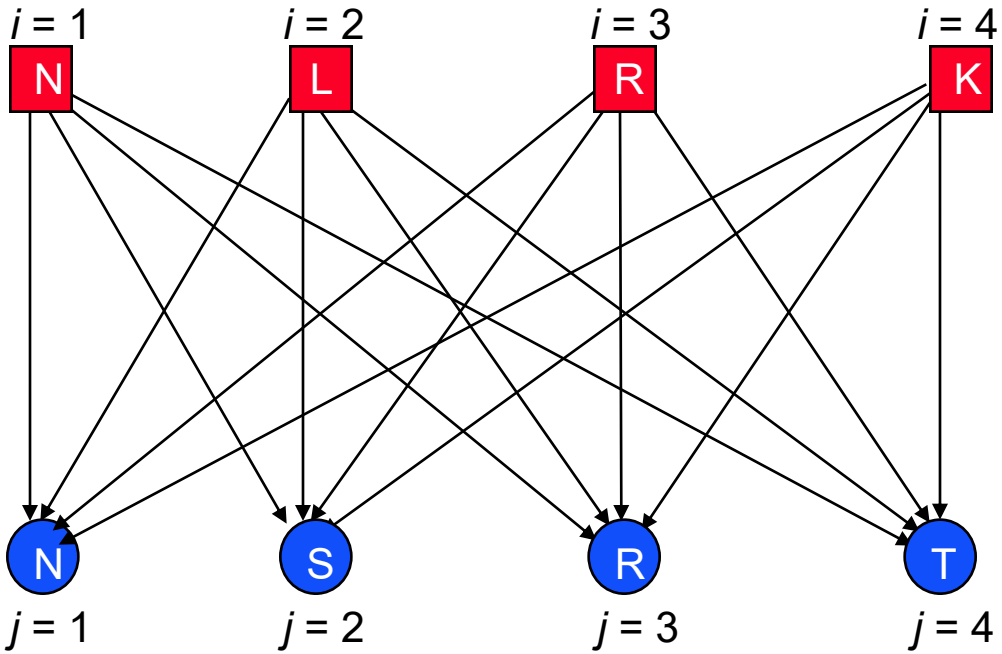
- Plant design, factor costs, and exchange rates affect ***production*** costs at each plant. ***Annual Fixed costs*** are incurred regardless of the production volume.

Plant Data	Unit Cost ($ per ton)	Fixed Cost ($000)	Capacity (tons/yr)
Newark	34,900	1,800	100
Los Angeles	32,200	2,750	200
Rotterdam	38,350	2,100	120
Kuala Lumpur	23,400	1,950	250

- ***Transportation costs*** include freight and tariffs

($ per ton) from plant\ to market	Newark	São Paulo	Rotterdam	Tokyo
Newark	--	12,225	9,075	21,450
Los Angeles	4,500	16,500	13,350	17,850
Rotterdam	9,150	12,600	--	12,525
Kuala Lumpur	21,450	18,450	15,150	5,925

Describe a Transportation Network



a_i = available capacity at plant i
(*e.g.*, a_2 = 200 tons)
c_i = unit cost at plant i
(*e.g.*, c_3 = \$38,350)

t_{ij} = unit transportation cost from plant i to market j
(*e.g.*, t_{33} = 0, t_{42} = 18,450)

d_j = demand in market j
(*e.g.*, d_1 = 150 tons)
p_j = unit price in market j
(*e.g.*, p_4 = \$62,650)

(The index i refers to plants, where 1 is Newark, 2 Los Angeles, 3 Rotterdam, and 4 Kuala Lumpur.
The index j refers to markets, where 1 is Newark, 2 São Paulo, 3 Rotterdam, and 4 Tokyo.)

Describe the problem in words

Problem Statement:
Maximize annual profit contribution by
determining the annual *production and transportation schedule*, ensuring that *plant capacity* and *market demand* limits are not exceeded.

What are the ***decision variables***?

Describe the problem in words

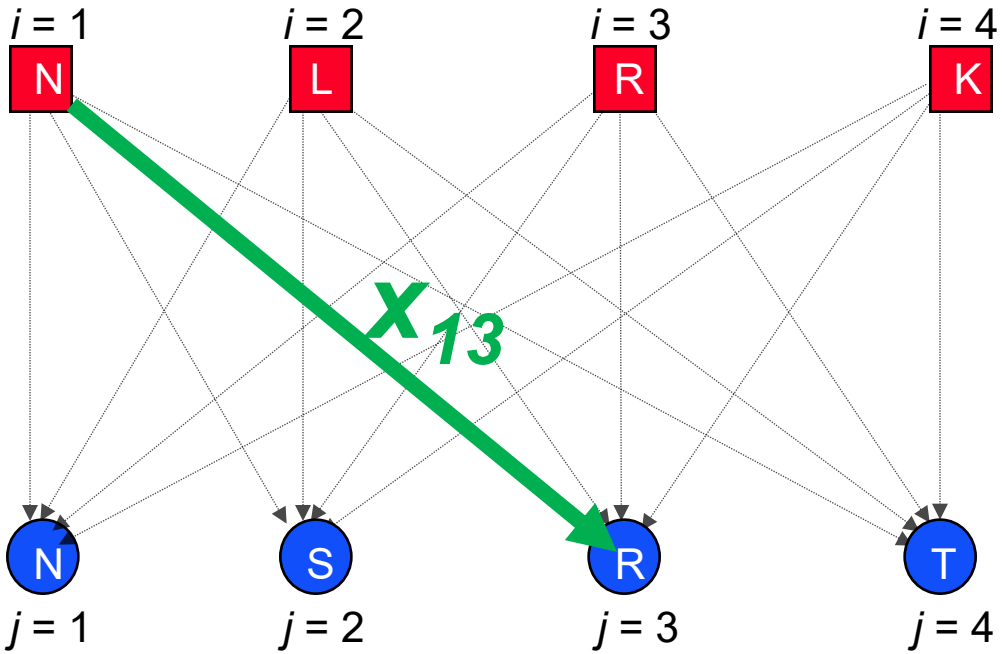
Problem Statement:
Maximize annual profit contribution by determining the annual *production and transportation schedule*, ensuring that *plant capacity* and *market demand* limits are not exceeded.

What are the ***decision variables***?

$$X_{ij}$$

Number of tons to produce in plant *i* that will be shipped to DC *j*

Decision variables



x_{ij} = units produced in plant i, and shipped to DC j

Need 4x4 = 16 decision variables

$(x_{11}, x_{12}, x_{13}, x_{14}$
$x_{21}, x_{22}, x_{23}, x_{24}$
$x_{31}, x_{32}, x_{33}, x_{34}$
$x_{41}, x_{42}, x_{43}, x_{44})$

(The index *i* refers to plants, where **1 is Newark**, 2 Los Angeles, 3 Rotterdam, and 4 Kuala Lumpur.
The index *j* refers to markets, where 1 is Newark, 2 São Paulo, **3 Rotterdam**, and 4 Tokyo.)

Constraints in Words

- Capacity Constraints: production at each plant must be less than capacity

(production at Newark plant) = ≤
(production at Los Angeles plant) = ≤
(production at Rotterdam plant) = ≤
(production at Kuala Lumpur plant) = ≤

- Demand Constraints: cannot ship more than the demand in any market

(total shipped to Newark mkt) = ≤
(total shipped to São Paulo mkt) = ≤
(total shipped to Rotterdam mkt) = ≤
(total shipped to Tokyo mkt) = ≤

- Non-negativity constraints: All shipment quantities must be nonnegative

Constraints in Words

- Capacity Constraints: production at each plant must be less than capacity

(production at Newark plant) =	$x_{11} + x_{12} + x_{13} + x_{14}$	$\leq c_1 = 100$ *tons*
(production at Los Angeles plant) =		$\leq$
(production at Rotterdam plant) =		$\leq$
(production at Kuala Lumpur plant) =		$\leq$

- Demand Constraints: cannot ship more than the demand in any market

(total shipped to Newark mkt) =	$\leq$
(total shipped to São Paulo mkt) =	$\leq$
(total shipped to Rotterdam mkt) =	$\leq$
(total shipped to Tokyo mkt) =	$\leq$

- Non-negativity constraints: All shipment quantities must be nonnegative

Constraints in Words

- Capacity Constraints: production at each plant must be less than capacity

(production at Newark plant) =	$x_{11} + x_{12} + x_{13} + x_{14}$	$\leq c_1 = 100$ *tons*
(production at Los Angeles plant) =		$\leq$
(production at Rotterdam plant) =		$\leq$
(production at Kuala Lumpur plant) =		$\leq$

- Demand Constraints: cannot ship more than the demand in any market

(total shipped to Newark mkt) =	$x_{11} + x_{21} + x_{31} + x_{41}$	$\leq d_1 = 150$ *tons*
(total shipped to São Paulo mkt) =		$\leq$
(total shipped to Rotterdam mkt) =		$\leq$
(total shipped to Tokyo mkt) =		$\leq$

- Non-negativity constraints: All shipment quantities must be nonnegative

LP Formulation of the Pfizer Problem (Contd.)

Constraints:

- *Production Capacity May Not Be Exceeded*

$$x_{11} + x_{12} + x_{13} + x_{14} \leq 100 \quad \text{(Capacity at Newark)}$$
$$x_{21} + x_{22} + x_{23} + x_{24} \leq 200 \quad \text{(Capacity at Los Angeles)}$$
$$x_{31} + x_{32} + x_{33} + x_{34} \leq 120 \quad \text{(Capacity at Rotterdam)}$$
$$x_{41} + x_{42} + x_{43} + x_{44} \leq 250 \quad \text{(Capacity at Kuala Lumpur)}$$

- *Maximum Demand May Not Be Exceeded*

$$x_{11} + x_{21} + x_{31} + x_{41} \leq 150 \quad \text{(Demand at Newark)}$$
$$x_{12} + x_{22} + x_{32} + x_{42} \leq 75 \quad \text{(Demand at São Paulo)}$$
$$x_{13} + x_{23} + x_{33} + x_{43} \leq 200 \quad \text{(Demand at Rotterdam)}$$
$$x_{14} + x_{24} + x_{34} + x_{44} \leq 100 \quad \text{(Demand at Tokyo)}$$

- *Non-Negativity*

$x_{ij} \geq 0$ for all plants i and markets j

Objective Function Formulation in Words

- ***Objective Function in words:*** maximize the annual profit contribution

Max *Profit Contribution = Revenues - Production Costs - Transportation Costs*

Revenues *= price (per unit) * number of units shipped to each DC j*

Production costs*= costs (per unit) * number of units produced in each plant i*

Transportation costs *= costs (per unit) from each plant i to each DC j * number of units produced in plant i and shipped to DC j*

Complete LP Formulation of the Pfizer Problem

Decision Variables:

x_{ij} = tons of product produced at plant *i* and shipped to market *j*

Objective Function:

Maximize profit contribution = *Revenue – Production Cost – Transportation Cost*

i.e., Max $55500\,(x_{11} + x_{21} + x_{31} + x_{41})$ (Revenue at Newark)

$+ \; 61100\,(x_{12} + x_{22} + x_{32} + x_{42})$ (Revenue at São Paulo)

$+ \; 57800\,(x_{13} + x_{23} + x_{33} + x_{43})$ (Revenue at Rotterdam)

$+ \; 62650\,(x_{14} + x_{24} + x_{34} + x_{44})$ (Revenue at Tokyo)

$- \; 34900\,(x_{11} + x_{12} + x_{13} + x_{14})$ (Production cost at Newark)

$- \; 32200\,(x_{21} + x_{22} + x_{23} + x_{24})$ (Production cost at Los Angeles)

$- \; 38350\,(x_{31} + x_{32} + x_{33} + x_{34})$ (Production cost at Rotterdam)

$- \; 23400\,(x_{41} + x_{42} + x_{43} + x_{44})$ (Production cost at Kuala Lumpur)

$- \; (0\,x_{11} + 12225\,x_{12} + 9075\,x_{13} + 21450\,x_{14} + 4500\,x_{21} + \ldots + 15150\,x_{43} + 5925\,x_{44})$ (Transportation cost)

(optional) For the mathematically inclined, in compact notation . . .

The following formulation represents the problem in a *generic* form; see the second example in the *Notes on Mathematical Notation* for a comparison.

As before, we use the indices *i* and *j for* plants and markets, receptively. *N* represents the number of plants and *M* the number of markets.

Let π_{ij} denote the unit profit *contribution* of the product produced at plant *i* and sold in market *j*.

$\pi_{ij} = p_j - c_i - t_{ij}$.

$$\max \quad \sum_{i=1}^{N}\sum_{j=1}^{M} \pi_{ij} x_{ij}$$

subject to:

$$\text{(capacity)} \quad \sum_{j=1}^{M} x_{ij} \le a_i \quad i = 1,\ldots,N$$

$$\text{(demand)} \quad \sum_{i=1}^{N} x_{ij} \le d_j \quad j = 1,\ldots,M$$

$$\text{(nonneg)} \quad x_{ij} \ge 0 \quad i = 1,\ldots,N, j = 1,\ldots,M$$

Problem Statement:
Maximize annual profit contribution by determining the *production and transportation schedule*, ensuring that *plant capacity* and *market demand* constraints are observed.

LP Solution to the Pfizer Problem

```
 -------------- RESULTS --------------

Optimal objective function value = 11616000.0
x11 100.0
x12 0.0
x13 0.0
x14 0.0
x21 50.0
x22 5.0
x23 0.0
x24 0.0
x31 0.0
x32 0.0
x33 120.0
x34 0.0
x41 0.0
x42 70.0
x43 80.0
x44 100.0
```

Use tabulate() function for better output:

From/To	Newark	SaoPaolo	Rotterdam	Tokyo
Newark	100	0	0	
Los Angeles	50	5	0	0
Rotterdam	0	0	120	0
Kuala Lumpur	0	70	80	100